

Module 1: Work Order Management

Tab 5: Excavation

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1. Purpose

This tab manages the entire on-site excavation process, from team assignment and internal permit generation to photo documentation and material reconciliation. It serves as the primary interface for the Project Manager, Site Manager, and Excavation Team Leader to coordinate and document all excavation-related activities.

2. Section 1: Team Assignment & Task Request

This section is the first step in the excavation process and is controlled by the Project Manager.

- **Team Selection:**
 - A multi-select dropdown labeled "Excavation Team(s)" will be populated with the predefined excavation teams from the HR Module.
 - The PM can select one or more teams for the assignment.
- **Task Assignment Request:**
 - Once the team is selected, a button labeled Generate & Send Task Request becomes active.
 - Clicking this button will:
 - 1.1.1 Auto-populate the "Form - Task Assignment Request" with information from the Work Order (Work Order No., Location, Work Description, etc.) and the assigned team leader's details (Name, Mobile, Iqama No.).
 - 1.1.2 Present the form in a modal view for the PM to review.
 - 1.1.3 Upon confirmation, the system will send the form (or a notification containing its data) to the designated SEC representative and consultant.
- **Task Number Entry:**
 - A field labeled "Task Number" will be available for the PM to enter the task number once it is received via SMS.
 - The rest of the tab remains locked until the Task Number is entered.

3. Section 2: SEC Internal Permits

This section is for managing the required internal SEC safety permits. It becomes active after the Task Number is entered.

- **Permit Generation:**
 - A button Generate SEC Permits will be available.
 - This will open a view with a list of required permits (e.g., Excavation Work (EXC), Limitation of Access (LOA), Work at Height (WAH), Lifting Operating (LOP)).
 - The system will auto-fill common information (Work Order No., Location, etc.) into each permit form.
 - The Issuer and Receiver (as designated in the Task Assignment Request) can then complete the remaining fields and provide digital signatures.
- **Permit Viewer:**
 - A document gallery will display all generated and signed SEC permits for this Work Order.

4. Section 3: Daily Work & Safety Forms

This section is for the daily management of on-site activities.

- **Daily Forms List:**
 - A table will list all daily forms generated for this excavation job, sorted by date.
 - Columns: Date, Form Type, Created By, Status.
 - A button Add Daily Forms will allow the creation of new daily forms.
- **Form Generation & Workflow:**
 - **Contractor Team Work Form:**
 - Generated daily. Auto-filled with the assigned team members.
 - The Site Manager or Team Leader can check a box next to any team member who is absent for that day.
 - **Safety Meeting Form:**
 - Generated daily. Auto-filled with project details.
 - This form is editable **only by users with the HSE role**. They can fill in the meeting topic and manage the attendee list.
 - **Daily Excavation Inspection Form:**
 - Generated daily. The Site Manager or a designated Competent Person completes the safety checklist.

5. Section 4: Photo Documentation

This section is for capturing and organizing all required site photos.

- **Photo Groups:** The interface will be organized into predefined groups:

1.2 Photos Prior to Site Safety Preparations

1.3 Photos After Site Preparations

1.4 Photos During Site Work

1.5 Photos After Excavation Work Completed

- **Upload & Capture:**

- Each group will have an Add Photos button that opens a file uploader or the device camera.
- All photos will have the required metadata automatically embedded (Date, Time, GPS Coordinates, Compass Direction).
- Photos will be automatically named according to the convention: [Request Number]-[Work Type]-[GroupName]-[Photo#].jpg.

- **Photo Gallery:** Each group will display a gallery of its uploaded photos.

6. Section 5: Material Reconciliation

This section is for tracking the actual materials used during excavation against the planned quantities.

- **Materials Table:**

- The table will list all materials from the "Work Order Information" tab.
- The table will have the following columns:

Column Header	Data Source / Control	Notes
Material Code	<u>work_orders.materials.item_code</u>	Read-only
Material Description	<u>work_orders.materials.description</u>	Read-only
Unit	<u>work_orders.materials.unit</u>	Read-only
Planned Qty	<u>work_orders.materials.estimated_quantity</u>	Read-only
Actual Qty Used	Number (Input)	Site Manager or Technician enters the actual quantity used.
Balance	Calculated Field	<u>Planned Qty - Actual Qty Used</u>

- **Generate Materials Balance Form Button:**

- After the Actual Qty Used is entered for all items, this button becomes active.
- Clicking it will auto-populate the "Form - Materials Balance" with the data from the table, ready for review and submission.

7. Section 6: One-Time Forms

This section is for forms that are typically filled out only once per excavation job.

- **Fossil Information Form (F2):**
 - A button Generate Fossil Information Form will auto-populate the form with project details, ready for completion if required.

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